

Algebra II Radicals and Rational Exponents Cheatsheet

Module 6

Rational Exponent Forms

$$\begin{aligned}a^{1/n} &= \sqrt[n]{a} \\a^{m/n} &= \sqrt[n]{a^m} = (\sqrt[n]{a})^m \\a^{-m/n} &= \frac{1}{a^{m/n}}\end{aligned}$$

Radical Rules

$$\begin{aligned}\sqrt[n]{ab} &= \sqrt[n]{a} \sqrt[n]{b} \\\sqrt[n]{\frac{a}{b}} &= \frac{\sqrt[n]{a}}{\sqrt[n]{b}} \quad (b \neq 0) \\\sqrt[n]{a^n} &= \begin{cases} |a|, & n \text{ even} \\ a, & n \text{ odd} \end{cases}\end{aligned}$$

Simplifying

Pull out perfect powers.

$$\sqrt[3]{54x^5} = \sqrt[3]{27x^3 \cdot 2x^2} = 3x\sqrt[3]{2x^2}$$

Combine only like radicals:

$$4\sqrt{7} - 9\sqrt{7} = -5\sqrt{7}$$

Rationalizing

Single radical denominator:

$$\frac{5}{\sqrt{3}} = \frac{5\sqrt{3}}{3}$$

Binomial radical denominator:

$$\frac{1}{a + \sqrt{b}} \cdot \frac{a - \sqrt{b}}{a - \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$$

Solving Radical Equations

1. Isolate one radical.
2. Raise both sides to the needed power.
3. Solve the resulting equation.
4. Check in the original equation.

Example template:

$$\sqrt{A} = B \Rightarrow A = B^2$$

with

$$B \geq 0$$

Extraneous Solutions

Squaring can introduce false solutions. Every solution must satisfy the original equation and any domain restrictions.

Quick Checks

- Even-index radicals require nonnegative radicands over the reals.
- Odd-index radicals can have negative radicands.
- Rational exponents follow exponent rules.
- Simplified radicals have no removable perfect-power factors.